

Normative [11C]PBR28 PET-MRI Pipeline: Neuroinflammation and Age

Overview

This project examines the relationship between age and regional neuroinflammation in healthy adults using [11C]PBR28 PET imaging. [11C]PBR28 is a TSPO-targeting PET tracer that can be used to investigate neuroinflammatory processes.

The project was developed as a reproducible PET-MRI analysis pipeline, covering the main stages of the analysis from data download and image preprocessing to regional PET quantification and statistical analysis. FSL was used for MRI preprocessing and spatial registration, while Python was used for ROI-based signal extraction, statistical testing, multiple-comparisons correction, and visualization.

Data

The data were obtained from the OpenNeuro dataset ds007907:

OpenNeuro: ds007907

The analysis was restricted to healthy-control participants. A total of 22 participants were included in the final analysis.

The dataset contains T1-weighted MRI and [11C]PBR28 PET images. PET images were used to estimate regional tracer uptake, expressed as standardized uptake values (SUV).

Analysis Pipeline

The analysis was organized into four main steps.

1. Selection and Download of Healthy-Control Data

Script: 01_get_hc_subjects.sh

This script identifies the healthy-control participants and downloads the required T1-weighted MRI and PET images for those participants. Restricting the analysis to healthy controls allows the project to focus on age-related variation without including patient-related effects.

2. MRI Preprocessing and PET Registration

Script: 02_preprocess_t1_pet.sh

The T1-weighted MRI images are first skull-stripped using FSL's BET tool. The anatomical images are then registered to the MNI152 standard space using linear registration with FLIRT followed by nonlinear registration with FNIRT.

The resulting transformations are applied to the PET images using `applywarp`, bringing the PET data into the same standard space. This allows regional PET measurements to be extracted consistently across participants.

3. ROI-Based PET Quantification

Script: `03_extract_roi_values.py`

Regional PET uptake is extracted using the Harvard-Oxford anatomical atlas. For each participant, the script calculates the mean PET SUV within 11 selected brain regions.

The resulting measurements are organized into a subject-level table containing demographic information, including age and sex, together with the regional PET values.

4. Statistical Analysis and Visualization

Script: `04_analyze_and_plot.py`

The relationship between age and regional PET uptake is assessed separately for each ROI.

Two types of correlation analyses are performed:

- Pearson correlation between age and regional SUV.
- Partial correlation between age and regional SUV while controlling for sex.

Because 11 brain regions are tested independently, multiple-comparisons correction is applied using the Benjamini-Hochberg False Discovery Rate (FDR) procedure.

The script also generates summary plots and scatter plots to visualize the observed relationships between age and regional PET uptake.

Results

After FDR correction across the 11 regional tests, none of the examined regions showed a statistically significant association between age and [11C]PBR28 SUV.

The Right Amygdala showed the strongest uncorrected association with age, with:

- Pearson correlation: **$r = 0.55$**
- Uncorrected p-value: **$p = 0.008$**
- FDR-adjusted p-value: **$p_{FDR} = 0.088$**

Therefore, the Right Amygdala association did not reach the predefined significance threshold after correction for multiple comparisons.

The result may be considered an exploratory finding that could be investigated further using a larger sample. Given the relatively small sample size of 22 healthy participants, the study has limited statistical power to detect modest age-related effects.

Why Multiple-Comparisons Correction Was Used

The analysis evaluates 11 brain regions independently. Performing multiple statistical tests increases the probability of obtaining a statistically significant result by chance.

To address this issue, the Benjamini-Hochberg FDR procedure was applied to the regional p-values. The corrected results were used to determine whether an observed age-SUV association remained statistically significant after accounting for the number of regions tested.

This approach prevents an isolated uncorrected p-value from being interpreted as strong evidence when multiple regional tests have been performed.

Software and Tools

The pipeline uses:

- **FSL** (≥ 6.0) for MRI preprocessing and spatial registration
- **Python** for data processing and statistical analysis
- **Nibabel** for working with NIfTI neuroimaging files
- **Nilearn** for atlas-based ROI analysis
- **Pandas** for data management
- **SciPy** for statistical analysis
- **Seaborn** for visualization

FSL must be installed and the \$FSLDIR environment variable must be correctly configured before running the preprocessing stage.

Pipeline Execution

The complete pipeline can be executed sequentially using the following commands:

```
bash scripts/01_get_hc_subjects.sh
```

```
bash scripts/02_preprocess_t1_pet.sh
```

```
python3 scripts/03_extract_roi_values.py
```

[python3 scripts/04_analyze_and_plot.py](#)

The first two scripts handle data preparation and image preprocessing, while the Python scripts perform ROI extraction and statistical analysis.

Output Files

The main outputs of the analysis include:

- `age_effect_summary.png` : summary visualization of the age effects across the examined brain regions.
- `scatter_right_amygdala.png` : scatter plot showing the relationship between age and Right Amygdala SUV.
- `age_suv_correlations.csv` : complete table containing the regional correlation results and corrected p-values.

Limitations

The main limitation of this analysis is the relatively small sample size ($n = 22$), which limits the ability to detect weaker age-related effects.

In addition, [^{11}C]PBR28 is a TSPO tracer, and interpretation of its signal as a direct measure of neuroinflammation requires caution. The analysis also does not include adjustment for the TSPO rs6971 polymorphism, which can affect [^{11}C]PBR28 binding. Therefore, the current results should be interpreted as an exploratory analysis of regional [^{11}C]PBR28 uptake rather than definitive evidence of age-related changes in neuroinflammation.

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